Recent Open Source Project Lab (ITL704)

MINI PROJECT REPORT ON

Casca

B. E. Information Technology

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CERTIFICATE

This is to certify that the project entitled Casca is a bonafide work of Keith Fernandes (221027), Charis Fernandes (221029), Joseph Kummari (221061), Aaron Vaz (221117) submitted to the University of Mumbai towards completion of mini project work for the subject of Recent Open Source Project Lab

(Name and sign)
Supervisor/Guide

(Dr. Prachi Raut)
HOD-IT

Examiners

1.----------

2.----------

Date:

DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Keith Fernandes	Charis Fernandes
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List of Abbreviations

Sr. No.	Abbrevation	Full Form
1	UX	User Experience
2	API	Application Programming Interface
3	CSS	Cascading Style Sheets
4	AI	Artificial Intelligence
5	DB	Database
6	UI	User Interface
7	AR	Augmented Reality

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CHAPTER 1
INTRODUCTION

1.1 Background

Casca is an innovative platform that empowers users to fully personalize their smartphone covers, offering complete creative control from design to final look. It focuses on transforming ordinary phone accessories into personalized style statements. Users can choose from a wide range of colors, patterns, and textures—or go a step further by uploading their own artwork or photographs to craft truly one-of-a-kind covers.

The platform is built using Next.js and React, ensuring a fast, dynamic, and SEO-friendly user experience. Prisma serves as the ORM layer, providing a seamless and type-safe connection with the database, while Tailwind CSS enables a clean, responsive, and modern design system that adapts effortlessly across devices. Together, these technologies enable smooth customization workflows, real-time previews, and efficient rendering of user-generated designs. Casca combines design freedom with cutting-edge web technologies, offering users not just a product—but an experience that bridges creativity, technology, and self-expression.

1.2 Scope of the Project

The project's scope is designed to cater to several user groups:

- Custom Case Buyers: Allows users to personalize phone cases, save their designs after logging in, and manage their user profiles.
- E-commerce Developers: Provides a modular system for building phone case platforms that include features like UPI payment integration, product reviews, and support for a dynamic catalog of phone models.
- Small Businesses & Vendors: The platform is suitable for sellers who offer custom products, though order handling remains a manual process.
- Order Trackers for Users: The system includes order tracking and cancellation capabilities, providing users with basic in-system status updates for their purchases.

1.3 Project Objectives

The main objectives of this project are:

1. To build a gallery page for other users to browse.
2. To add user profile and account settings management.
3. To enable order history, tracking, and cancellation functionalities.
4. To add feedback system.
5. To add a product review system.
6. To integrate product wishlist.
7. To allow for email notifications on logins.
8. To implement admin dashboard with graphical statistical representations

1.4 Problem Statement

Most e-commerce platforms for phone cases provide limited customization options and lack essential features such as order tracking, user profiles, and support for modern tools like AI-generated designs or the ability to save design drafts. This leads to a generic user experience with minimal personalization

CHAPTER 2
LITERATURE REVIEW

[1] Casca Custom Phone Case Design E-commerce Platform (2024):-
This project presents a fully functional MVP developed by students that allows users to design personalized phone cases. It integrates modern technologies like Next.js, React, and Prisma, and includes AI-based design recommendations along with Stripe for payments. User testing validated the platform's usability and customization features. However, it lacks key functionalities like user profiles, saved designs, order history, and admin dashboard.

[2] Zapvi (2018)
Zapvi is an established commercial website that offers custom phone case printing through an intuitive interface. The review analyzed its features, focusing on the design upload process, checkout flow, and compatibility across devices. The site employs a simple image upload system with limited drag-and-drop customization and supports payments through Razorpay and Paytm. Despite being user-friendly, it misses advanced personalization features and structured customer feedback mechanisms.

[3] CaseKaro (2024)
CaseKaro functions as an online store offering predesigned and custom phone covers through a traditional e-commerce setup. The review assessed user experience, product variety, and checkout performance. Built using PHP/Shopify/JS, the site relies on static product listings with limited customization flexibility. The absence of modern e-commerce features like order history, product reviews, saved designs, and UPI support highlights areas for improvement.

Sr No.	Title	Year of publication	Paper/Repository review	Research gaps identified
1	Casca Custom Phone Case Design E-commerce Platform	2024	A fully functional MVP was built by students, which included user testing, AI integration, and Stripe checkout.	Lack of user profile management, no integrated order history, absence of product reviews, no feedback system, no option to save designs, missing support for popular payment options like UPI, no cookie consent mechanism, and a lack of a modular phone model management system.
2	Zapvi	2018	The review involved a feature and UX analysis of the live site, evaluating customization depth, checkout flow, and device support. It uses a basic	No structured feedback or complaint submission system, no option to save designs or resume customization, and a lack of a modular phone model

			image upload system with a limited drag-and-drop UI and supports payments via Razorpay/Paytm	management system.
3	CaseKaro	2024	An observational analysis was conducted on the user experience, customization options, checkout process, and product categories. The platform uses static product listings with minimal customization and a traditional e-commerce stack (PHP/Shopify/JS).	No integrated order history, absence of product reviews, no structured feedback system, no option to save designs, missing support for UPI payments, no cookie consent mechanism, and lacks a modular system for phone model management.

Table 1: Literature Review

CHAPTER 3
SYSTEM DESCRIPTION

3.1 System block diagram

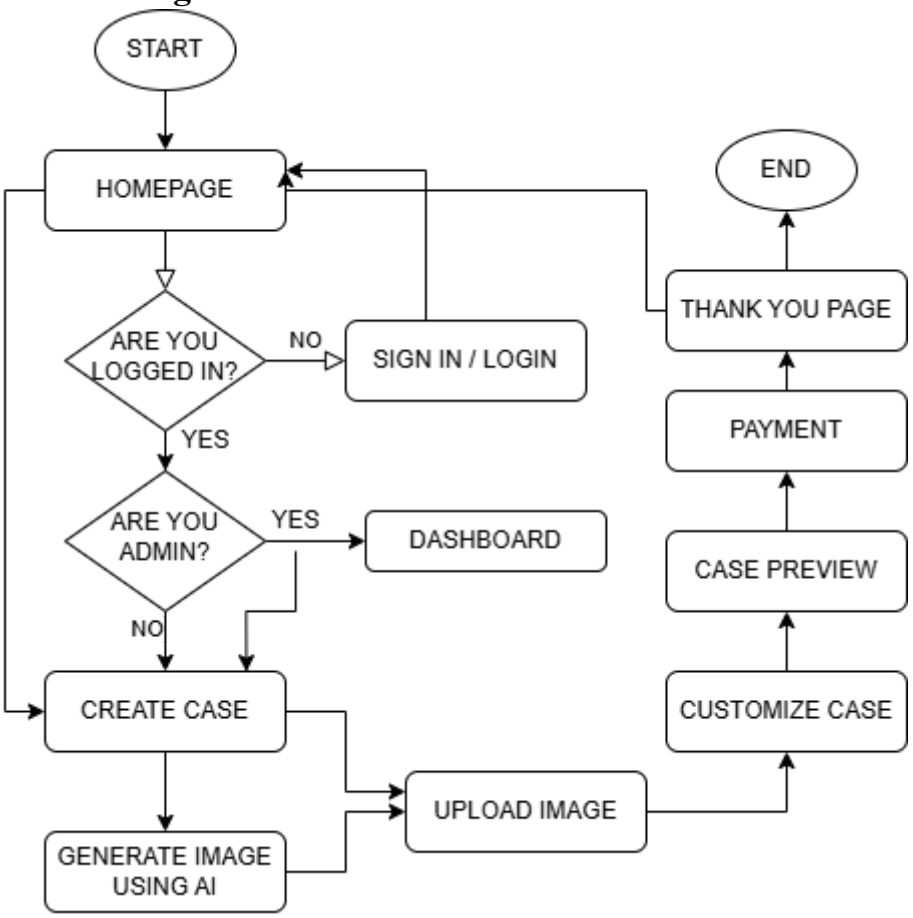

Fig 3.1:- System Block Diagram

3.2 Architectural details

This document outlines the system design for a website that enables users to create and purchase customizable phone covers. The flowchart (Figure 3.1) illustrates the primary components and interactions, depicting the user journey from accessing the homepage to completing a purchase.

Start: The process begins at the Start node, marking the user's entry into the system.

Homepage: Users are directed to the Homepage, showcasing the site's features and options to create customized covers.

Authentication Check: The system verifies if the user is logged in.

- **Not Logged In:** Users are directed to the Sign In/Login page.
- **Logged In:** The system performs an Admin Check.

Admin Check: If the user has admin privileges, they are directed to the Dashboard for admin functions. Non-admin users continue to the case creation process.

Case Creation Process: Non-admin users can:

- **Create Case:** Begin designing a custom cover.
- **Generate Image Using AI:** Use an AI tool to create a unique image.
- **Upload Image:** Upload a personal image for customization.

Customize Case: Users adjust elements, add personal touches, and finalize their design.

Case Preview: Users visualize the final product before proceeding.

Payment: Upon finalizing the design, users complete the purchase on the Payment page.

Thank You Page: After payment, users see a Thank You Page, marking the end of the purchase journey. Confirmation details are emailed.

End: The flow concludes at the End node.

Admin Functionality: The Dashboard, accessible only to admins, includes features like order management, user activity oversight, and content maintenance.

CHAPTER 4
TIMELINE CHART

Project Name :	Casca
Project Start:	14-07-2025
Project End:	27/10/2025
Weeks:	Wk 14

Task	Start Date	End Date	Days	Progess
Project Initiation Phase				
Group formation, leader appointment, topic search	14-07-2025	20-07-2025	6	100%
Topic finalization and proposal preparation	21-07-2025	27-07-2025	6	100%
Submission of project proposal	28-07-2025	03-08-2025	6	100%
Proposal presentation and approval	04-08-2025	10-08-2025	6	100%
Research and Planning Phase				
Literature review and logbook checking	11-08-2025	17-08-2025	6	100%
Implementation 1 backend and frontend structure	18-08-2025	24-08-2025	6	100%
Mid-semester evaluation and logbook verification	25-08-2025	31-08-2025	6	100%
Development Phase				
API and database setup	15-09-2025	21-09-2025	6	100%
Core feature implementation and testing	22-09-2025	28-09-2025	6	100%
Testing and Documentation Phase				
Feature demonstration, documentation	29-09-2025	05-10-2025	6	100%
Implementation improvement, final testing	06-10-2025	12-10-2025	6	100%
Finalization and Deployment Phase				
Project demonstration, report writing	13-10-2025	19-10-2025	6	100%
End-semester, final report and logbook submission	20-10-2025	27-10-2025	6	100%

4.1 Timeline Chart of the Project

CHAPTER 5
IMPLEMENTATION

5.1 Database Details

Firebase is a cloud-based platform by Google that provides backend services such as authentication, real-time databases, hosting, and storage, allowing developers to build and scale web or mobile applications efficiently without managing their own servers.

- Real-time Database & Storage: Used Firebase Firestore to store user data, designs, and order records securely with real-time synchronization.
- Authentication Support: Integrated Firebase Auth for easy and secure user login and session management.
- Hosting & Scalability: Leveraged Firebase Hosting for fast, reliable deployment and automatic scalability of the web app.

users (Collection)

- {documentId} **(Document)**
  - address**: (String)**
  - email**: (String)**
  - fullName**: (String)**
  - phone**: (String)**
  - updatedAt: **(Timestamp)**

5.2 Working of the project

This document outlines the system design for a website that enables users to create and purchase customizable phone covers.

Start: The process begins at the Start node, marking the user's entry into the system.

Homepage: Users are directed to the Homepage, showcasing the site's features and options to create customized covers.

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End: The flow concludes at the End node.

Admin Functionality: The Dashboard, accessible only to admins, includes features like order management, user activity oversight, and content maintenance.

5.3 Code

Layout.tsx

```
import type { Metadata } from "next";
import { Inter, Recursive } from "next/font/google";
import "./globals.css";
import Navbar from "@/components/Navbar";
import Footer from "@/components/Footer";
import Providers from "@/components/Providers";
import { Layout } from "lucide-react";
import Chatbot from "./configure/chatbot/page";
import GoogleTranslate from "@/components/GoogleTranslate";
import Script from "next/script";
import CookieBanner from "@/components/CookieBanner";

const recursive = Recursive({ subsets: ["latin"] });

export const metadata: Metadata = {
  title: "Create Next App",
  description: "Generated by create next app",
};

export default function RootLayout({
  children,
}: Readonly<{
```

```
  children: React.ReactNode;
}>) {
  return (
    <html lang="en">
      <body className={recursive.className}>
        <div className="h-12 w-12 z-50 fixed right-4 bottom-16">
          <Chatbot />
        </div>
        <Navbar />
        <main className="flex grainy-light flex-col min-h-[calc(100vh-3.5rem-1px)]">
          <div className="flex-1 flex flex-col h-full">
            <Providers>{children}</Providers>
          </div>
          <Footer />
        </main>
        <CookieBanner />
        <GoogleTranslate />
        <Script
          src="//translate.google.com/translate_a/element.js?cb=googleTranslateElementInit"
          strategy="afterInteractive"
        />
      </body>
    </html>
  );
}
```

firebase.ts

```
import { initializeApp } from "firebase/app";
import { getFirestore } from "firebase/firestore";

const firebaseConfig = {
  apiKey: "AIzaSyAxlo5ONJ5hIhTxhoy5dEhrYgSvjz7D-bw",
  authDomain: "casca-6b5fa.firebaseapp.com",
  projectId: "casca-6b5fa",
  storageBucket: "casca-6b5fa.firebasestorage.app",
  messagingSenderId: "106777504209",
  appId: "1:106777504209:web:e0ffa74443e04db986d8f0"
};

// Initialize Firebase
const app = initializeApp(firebaseConfig);
const db = getFirestore(app);

export { db };
```

CHAPTER 6
RESULTS AND DISCUSSIONS

6.1 Implementation screenshots

Fig 6.1.1 shows the gallery page to the customer from where they can choose an pre designed cover having 2 options either to buy directly or add to cart .

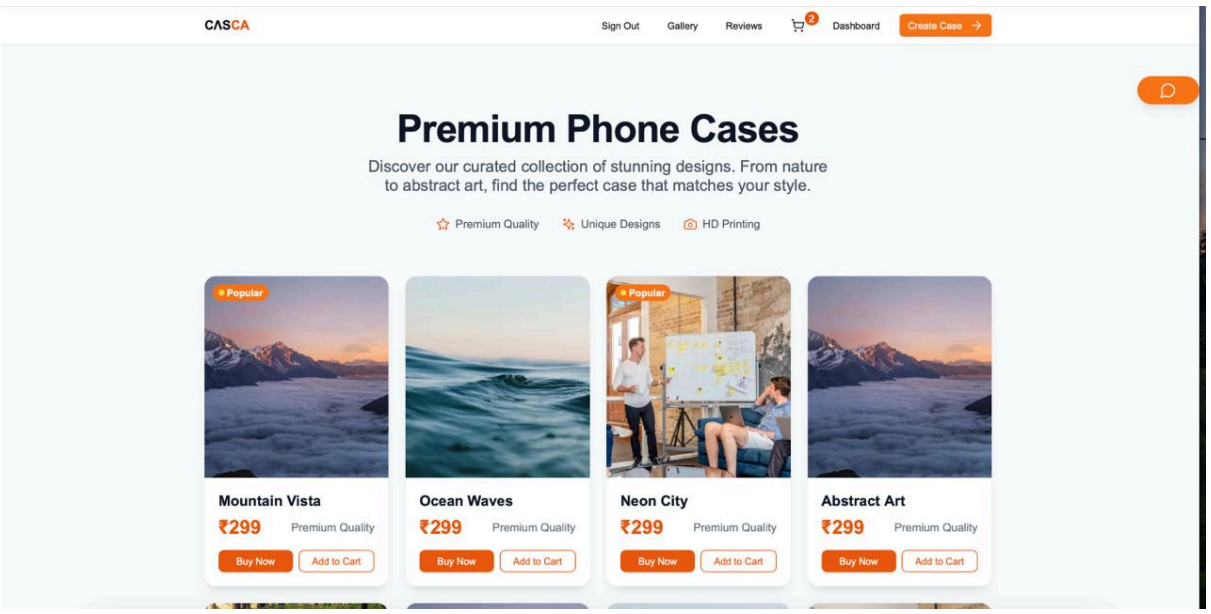

Fig 6.1.1 Gallery Page

Fig 6.1.2 is the cart page where your pre designed covers are previewed for the users to cross check also shows the order summary .

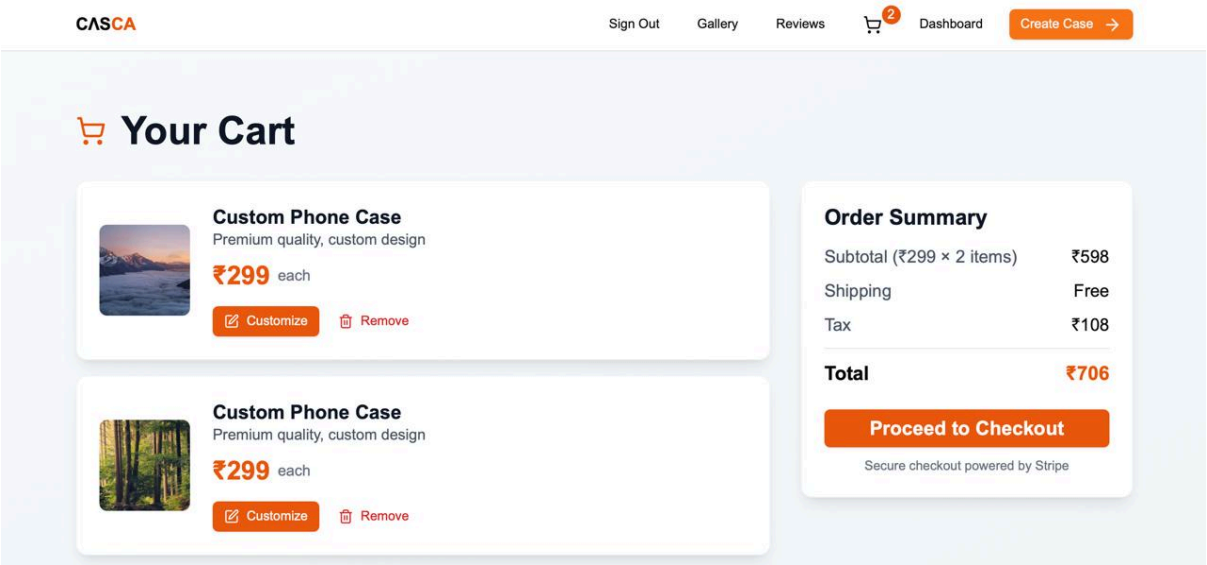

Fig 6.1.2 Cart Page

Fig 6.1.3 is the My Order page that shows the purchase history of the user and the product details.

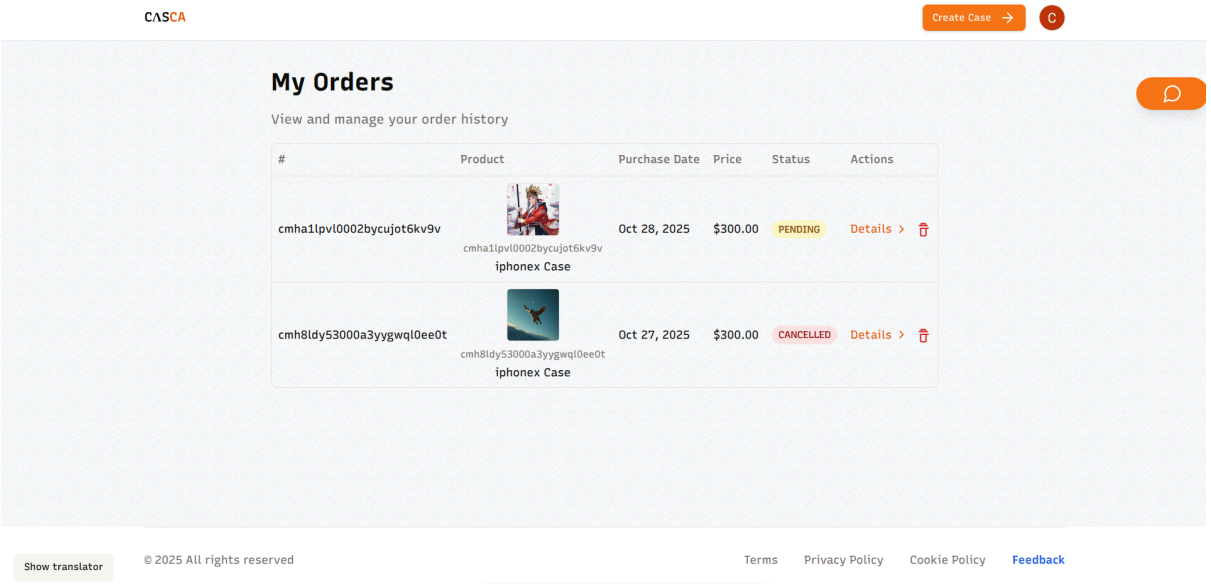

Fig 6.1.3 My orders Page

Fig 6.1.4 is My Profile page that shows the user information, user can edit his name, change password and add address for delivery.

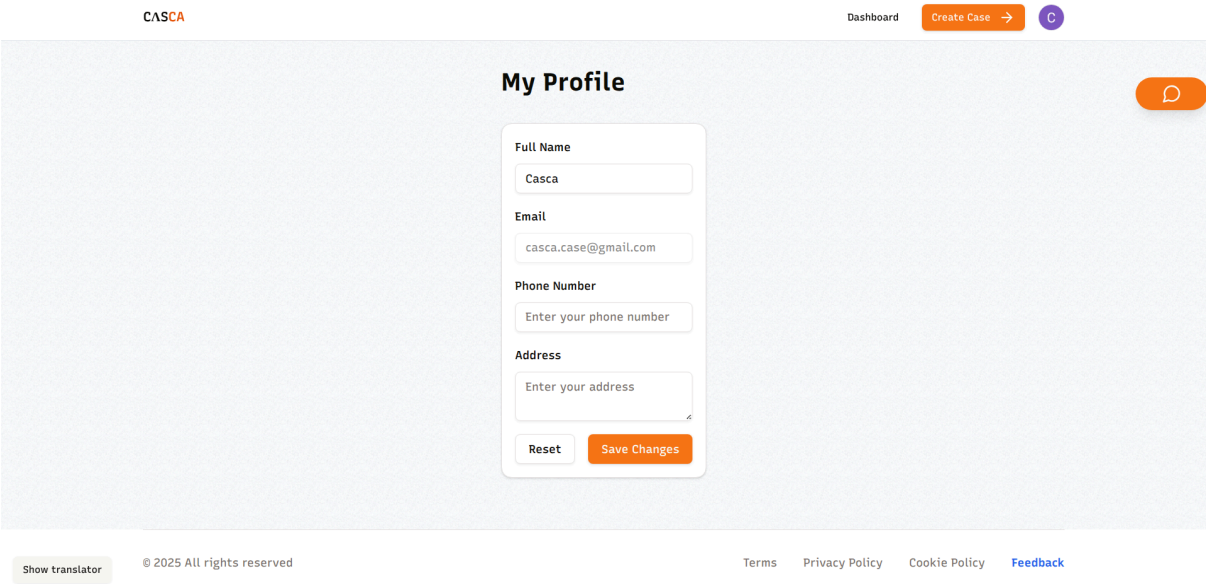

Fig 6.1.4 My Profile Page

Fig 6.1.5 is My Feedback page displays allows users to submit feedback including ratings, messages, and images for product improvement.

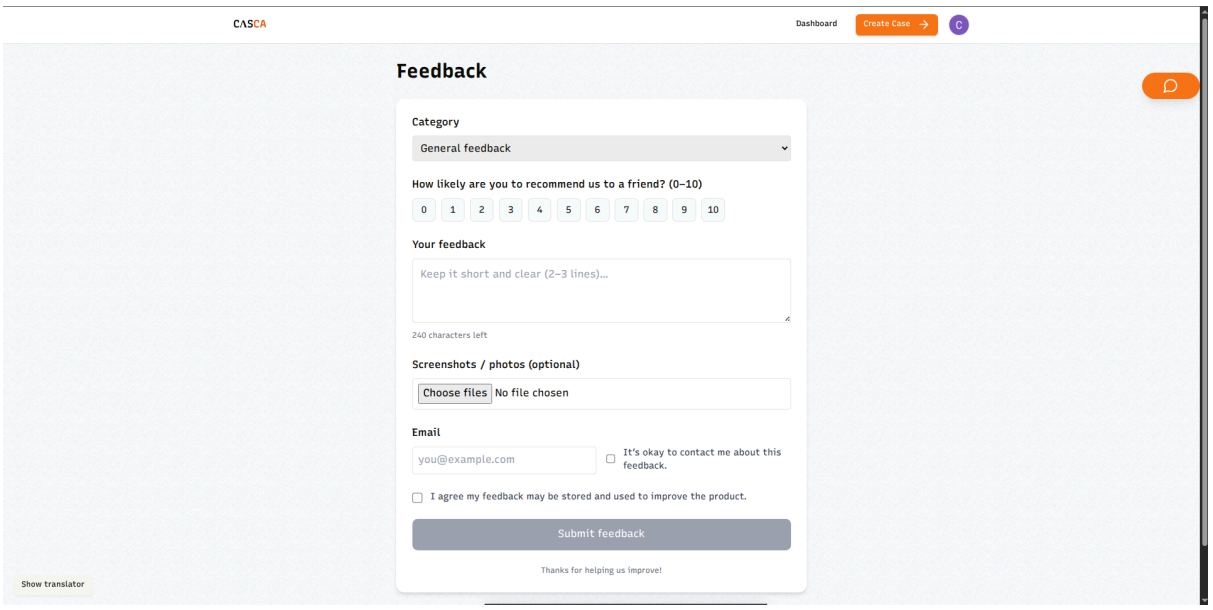

Fig 6.1.5 Feedback Page

Fig 6.1.6 is My Wishlist page displays saved custom phone case designs with images from Firebase, allowing add/remove functionality.

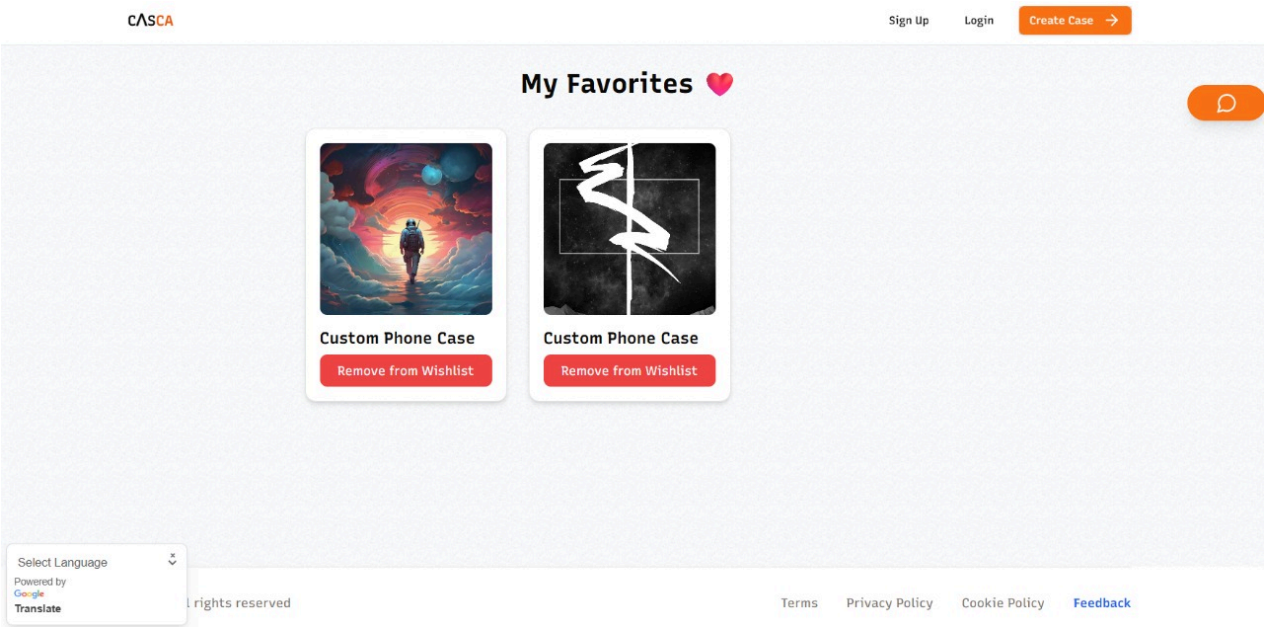

Fig 6.1.4 Wishlist Page

Fig 6.1.5 shows the admin dashboard , which visualises and summarizes all the order history and also the trends by which the customer orders the cases .

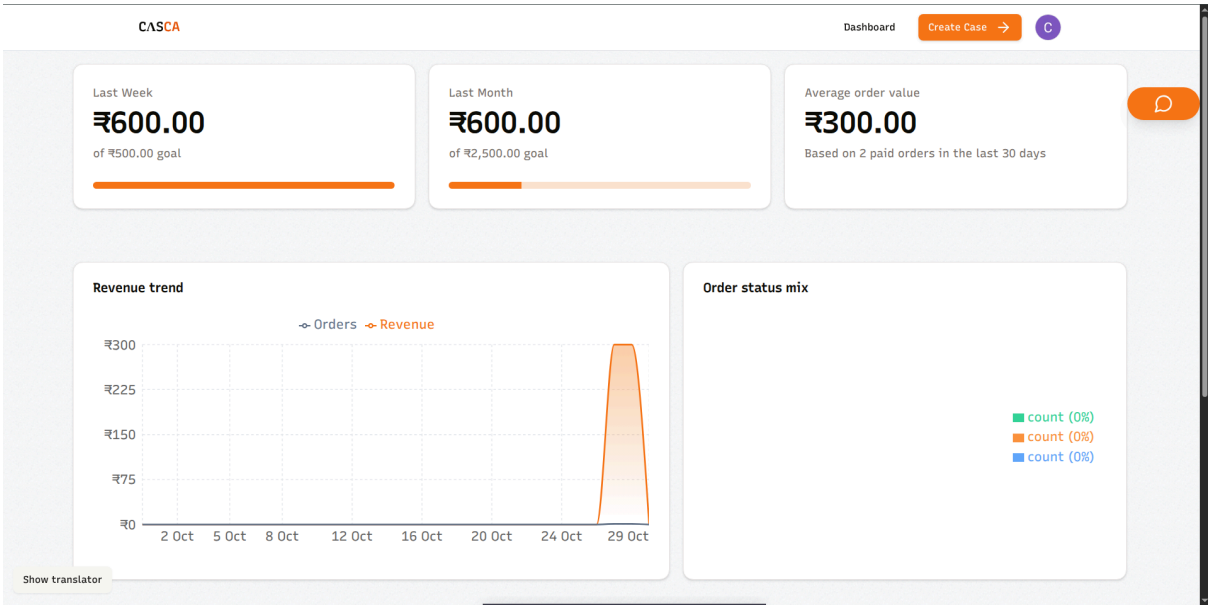

Fig 6.1.5 Admin Dashboard

Fig 6.1.6 shows the email interface, Which sends an automated email to users upon successful login, confirming their account access with details like login time and device information.

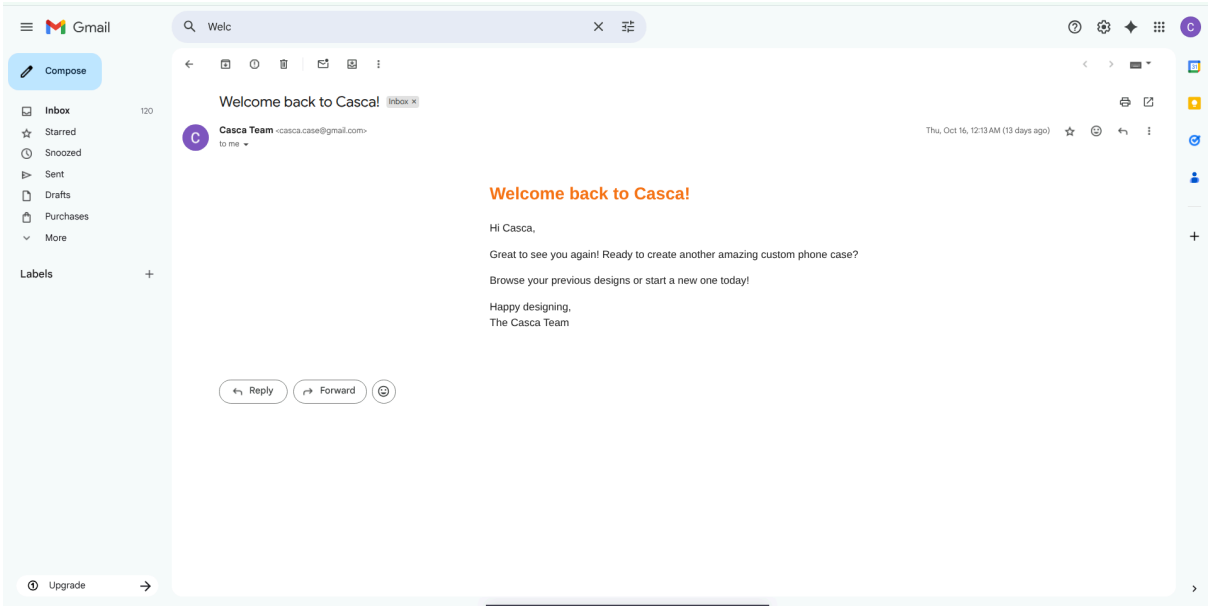

Fig 6.1.6 Email Interface

6.2 Result analysis

The implemented features significantly enhanced Casca's overall functionality and user experience. The addition of profiles, order tracking, reviews, feedback, and wishlists made the platform more interactive and user-centric, while email notifications and an admin dashboard improved management and security. Together, these upgrades transformed Casca from a basic prototype into a more complete, reliable, and scalable e-commerce solution.

1. Gallery Page: Enables users to browse and view case designs created by others, improving community interaction.

2. User Profile & Account Settings: Allows users to manage personal details, saved designs, and preferences easily.

3. Order History, Tracking & Cancellation: Provides users with complete control over their past and ongoing orders.

4. Feedback System: Collects user opinions and suggestions to enhance platform quality and usability.

5. Product Review System: Lets users rate and review products, increasing transparency and trust.

6. Wishlist Integration: Allows users to save favorite products for future purchase or reference.

7. Email Notifications: Sends automated alerts for logins and important account activities to improve security.

8. Admin Dashboard: Displays visual analytics of orders, users, and feedback through interactive charts and graphs.

CHAPTER 7
CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The Casca project represents a significant evolution of an open-source customizable phone case e-commerce platform, addressing the key limitations of the original version by introducing critical features that enhance both functionality and user experience. Through the integration of modern technologies such as Next.js, Prisma, and Stripe, the platform now supports user profile management, order tracking and cancellation, Email Notifications, product reviews and feedback, profile, and gallery support. These additions transform Casca from a minimal MVP into a fully functional, scalable, and user-oriented system. The platform now enables a seamless and personalized customer journey—from account creation and design customization to secure payment and order management—while maintaining the simplicity and open-source spirit of the original. This project not only improves usability and reliability but also lays the groundwork for future innovation, including AI-assisted design tools, analytics dashboards, AR-based previews, and deeper vendor integration, positioning Casca as a strong foundation for real-world commercial and educational applications.

7.2 Future Scope

AI-Powered Design Assistance – Integrate AI tools to suggest designs, color themes, or layouts based on user preferences.

Augmented Reality (AR) Preview – Allow users to visualize the case on their phone model in real time.

Referral and Reward System – Encourage user retention through loyalty points and friend referrals.

Print-on-Demand Integration – Automate design transfer to partner vendors for real-world printing.

Advanced Security and Privacy Controls – Implement cookie-based consent management and stronger data protection features.

REFERENCES

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2. M. Patel, "Zapvi." Zapvi, 2018. [Online]. Available: https://zapvi.in [Accessed: Aug. 4, 2025].

3. R. Parmar, "CaseKaro." Case Karo, Jul. 31, 2024. [Online]. Available: https://casekaro.com [Accessed: Aug. 4, 2025].

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We are thankful to our college **St. Francis Institute of Technology** for giving us this chance to gain exposure in solving real world problems and acquire practical knowledge and skill sets that will prove to be very crucial to our long-term career prospects. We would take this opportunity to express our sincerest gratitude to our esteemed director **Bro. Shantilal Kujur**, our Principal **Dr. Deepak Jayaswal** and our HOD, **Dr. Prachi Raut** for their encouragement, the direction that they give to our college and us students, and the facilities provided to us.
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